Step 1 : **alice** registered him as a user (client side)

```
Commands: register | login | send | inbox | logout | status | quit
>>> register
Username: alice
Password: alice
Set key password (or press Enter for none): check
✓ Generated new key pair for alice
✓ Registration successful: registered
```

Step 2 : **bob** registered him as a user (client side)

```
Commands: register | login | send | inbox | logout | status | quit
>>> register
Username: bob
Password: bob123
Set key password (or press Enter for none): check
✓ Generated new key pair for bob
✓ Registration successful: registered
```

Step 3 : **alice** and **bob** both registered successfully (server-side)

```
[SERVER] User registered: alice
127.0.0.1 - - [23/Feb/2026 00:02:40] "POST /register HTTP/1.1" 200 -
[SERVER] User registered: bob
127.0.0.1 - - [23/Feb/2026 00:03:24] "POST /register HTTP/1.1" 200 -
127.0.0.1 - - [23/Feb/2026 00:03:35] "POST /login HTTP/1.1" 401 -
```

Step 4 : **alice** gets login and then send message to bob (client-side)

```
Commands: register | login | send | inbox | logout | status | quit
>>> login
Username: alice
Password: alice
Enter key password (or press Enter for none): check
✓ Loaded existing keys for alice
✓ Login successful
  Session token: l3IgIZCWfUbFcF__o4gH...

Commands: register | login | send | inbox | logout | status | quit
>>> send
To: bob
Message: this is message from alice to bob
✓ Encrypted message sent to bob
  [Your plaintext was: 'this is message from alice to bob']
  [Server sees only: rO4xd42rxUyadf0eABV/0Xs4e/6pjoFfzmW7OSnMangsLq2zaZ...]
```

Step 5 : **alice** successfully gets login and successfully have send message to **bob** (server-side)

```
[SERVER] User logged in: alice
127.0.0.1 - - [23/Feb/2026 00:04:01] "POST /login HTTP/1.1" 200 -
127.0.0.1 - - [23/Feb/2026 00:04:15] "GET /public_key/bob HTTP/1.1" 200 -
[SERVER] Encrypted message stored (from alice to bob)
[SERVER] Ciphertext preview: rO4xd42rxUyadf0eABV/0Xs4e/6pjoFfzmW7OSnMangsLq2zaZ...
127.0.0.1 - - [23/Feb/2026 00:04:15] "POST /send HTTP/1.1" 200 -
```

Step 6 : now **bob** successfully login and checked his inbox (client-side)

```
Commands: register | login | send | inbox | logout | status | quit
>>> login
Username: bob
Password: bob123
Enter key password (or press Enter for none): check
✓ Loaded existing keys for bob
✓ Login successful
  Session token: QS5-jJ8VxyHF3Sv6wqXW...

Commands: register | login | send | inbox | logout | status | quit
>>> inbox

============================================================
📬 You have 1 message(s)
============================================================

Message 1:
  From: alice
  Content: this is message from alice to bob
  [Encrypted form: rO4xd42rxUyadf0eABV/0Xs4e/6pjoFfzmW7OSnM...]
```

You can see that **bob** gets the message from **alice** successfully

Step 7 : now **bob** successfully login and checked his inbox (server-side)

```
[SERVER] User logged in: bob
127.0.0.1 - - [23/Feb/2026 00:08:54] "POST /login HTTP/1.1" 200 -
[SERVER] User bob fetched 1 message(s)
127.0.0.1 - - [23/Feb/2026 00:08:57] "GET /messages HTTP/1.1" 200 -
```

Step 8: Now there is one more user created named “**charlie**” and **charlie** login and checked his inbox and his inbox is empty means he does not see the messaging between **alice** and **bob**

```
Commands: register | login | send | inbox | logout | status | quit
>>> register
Username: charlie
Password: charlie123
Set key password (or press Enter for none): check
✓ Generated new key pair for charlie
✓ Registration successful: registered

Commands: register | login | send | inbox | logout | status | quit
>>> login
Username: charlie
Password: charlie123
Enter key password (or press Enter for none): check
✓ Loaded existing keys for charlie
✓ Login successful
  Session token: otZ5GT312J8SEePNTfvx...

Commands: register | login | send | inbox | logout | status | quit
>>> inbox
📭 No messages
```

Step 9 : **charlie** login and inbox checking (server-side)

```
[SERVER] User registered: charlie
127.0.0.1 - - [23/Feb/2026 00:10:58] "POST /register HTTP/1.1" 200 -
[SERVER] User logged in: charlie
127.0.0.1 - - [23/Feb/2026 00:11:07] "POST /login HTTP/1.1" 200 -
[SERVER] User charlie fetched 0 message(s)
127.0.0.1 - - [23/Feb/2026 00:11:11] "GET /messages HTTP/1.1" 200 -
```